Paper source

Repository source: paper/README.md

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Release: v0.1.0 (September 6, 2026)
Canonical repository: https://github.com/swihart/minvol-degree3-spectral-bound
Versioned release: https://github.com/swihart/minvol-degree3-spectral-bound/releases/tag/v0.1.0

The manuscript source is `degree3_spectral_refinement.tex`. To keep the build self-contained on minimal TeX installations, the short bibliography is included inside the TeX source.

Build from the repository root with:

```
./build_paper.sh
```

or directly inside this directory with:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error degree3_spectral_refinement.tex
```

The tracked PDF is `degree3_spectral_refinement.pdf`. The visible front matter and PDF metadata identify this manuscript as release `v0.1.0`, dated September 5, 2026.

Supporting proof documentation

For a deeper audit of the two nonroutine parts of the argument, see:

- `../proof/DEGREE3_TENSOR_IDENTITY.md` for the expanded cubic-tensor and quartic-moment derivation; and
- `../proof/BRIDGE_LEMMAS.md` for the expanded projection, nuclear-norm, duality, and smooth-approximation steps.

Typeset PDF versions are available under `../rendered/markdown/proof/`.

AI-assistance declaration

The manuscript identifies the system used as OpenAI ChatGPT (GPT-5.6 Sol Pro), accessed August-September 2026, and distinguishes computational reproducibility from independent mathematical review. The fuller repository-level account is in `../AI_ASSISTANCE.md`.

Status language

Use:

> AI-assisted, non-peer-reviewed research draft seeking independent mathematical verification.

Do not describe the draft as peer reviewed, certified, or independently verified.

The repository-wide status and reproduction instructions are maintained in `../VERIFICATION_STATUS.md` and `../REPRODUCIBILITY.md`.